GitOps for DynaKube

Series: K8S | **Notebook:** 3 of 12 | **Created:** January 2026 | **Last Updated:** 01/30/2026

Managing DynaKube with ArgoCD and Flux

GitOps enables declarative, version-controlled management of your Dynatrace monitoring configuration. This notebook covers integrating DynaKube with popular GitOps tools: ArgoCD and Flux.

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Prerequisites

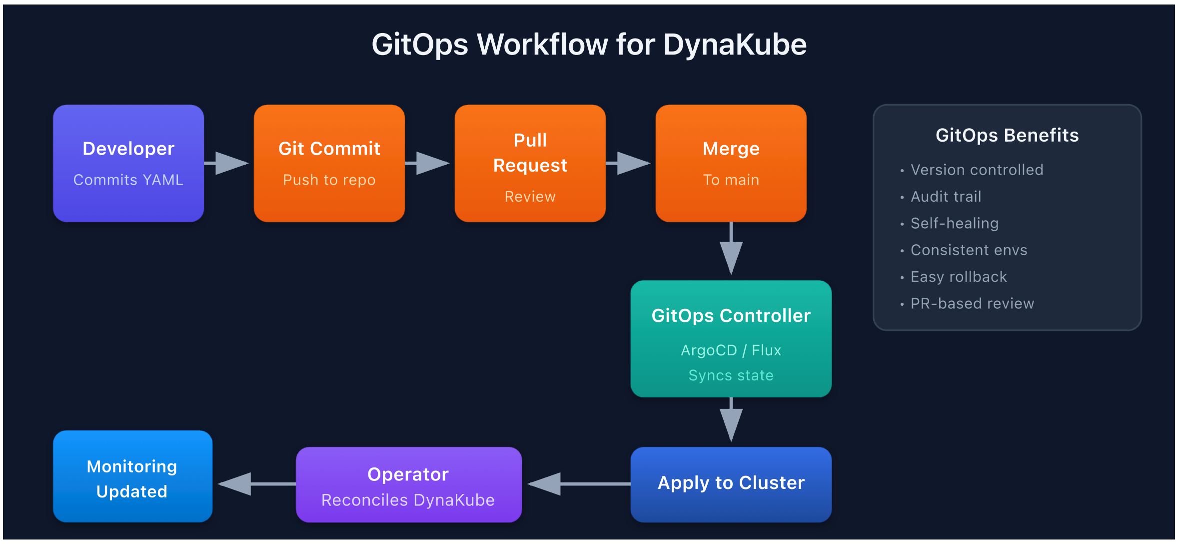
Requirement	Details
GitOps Tool	ArgoCD v2.x or Flux v2.x
Git Repository	Write access for configuration
Kubernetes	Dynatrace Operator installed
Knowledge	Familiarity with K8S-02: DynaKube Deployment

1. GitOps Principles for Observability

Why GitOps for Monitoring?

Benefit	Description
Version Control	Track all config changes in Git history
Audit Trail	Who changed what, when, and why
Consistency	Same config across dev/staging/prod
Self-Healing	Automatic drift correction
Rollback	Revert by reverting Git commits

GitOps Workflow for DynaKube



Repository Structure

```
monitoring-config/
├── base/
│   ├── namespace.yaml
│   ├── operator/
│   │   └── kustomization.yaml
│   └── dynakube/
│       ├── dynakube.yaml
│       └── kustomization.yaml
├── overlays/
│   ├── dev/
│   │   ├── kustomization.yaml
│   │   └── dynakube-patch.yaml
│   ├── staging/
│   │   └── ...
│   └── prod/
│       └── ...
└── clusters/
    ├── dev-cluster/
    ├── staging-cluster/
    └── prod-cluster/
```

2. ArgoCD Integration

ArgoCD Application for Operator

```yaml
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: dynatrace-operator
  namespace: argocd
spec:
  project: monitoring
  source:
    repoURL: https://raw.githubusercontent.com/Dynatrace/dynatrace-
operator/main/config/helm/repos/stable
    chart: dynatrace-operator
    targetRevision: 1.2.0  # Pin version for stability
    helm:
      releaseName: dynatrace-operator
      values: |
        platform: kubernetes
        csidriver:
          enabled: true
  destination:
    server: https://kubernetes.default.svc
    namespace: dynatrace
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
    syncOptions:
      - CreateNamespace=true
```

ArgoCD Application for DynaKube

```yaml
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: dynakube
  namespace: argocd
spec:
  project: monitoring
  source:
    repoURL: https://github.com/your-org/monitoring-config
    targetRevision: main
    path: overlays/prod
  destination:
    server: https://kubernetes.default.svc
    namespace: dynatrace
  syncPolicy:
    automated:
      prune: false  # Don't auto-delete DynaKube
      selfHeal: true
```

ArgoCD App of Apps Pattern

Manage multiple clusters from a single Application:

```yaml
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: monitoring-apps
  namespace: argocd
spec:
  project: monitoring
  source:
    repoURL: https://github.com/your-org/monitoring-config
    targetRevision: main
    path: clusters
  destination:
    server: https://kubernetes.default.svc
    namespace: argocd
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
```

3. Flux Integration

Flux HelmRelease for Operator

```yaml
apiVersion: source.toolkit.fluxcd.io/v1beta2
kind: HelmRepository
metadata:
  name: dynatrace
  namespace: flux-system
spec:
  interval: 1h
  url: https://raw.githubusercontent.com/Dynatrace/dynatrace-
operator/main/config/helm/repos/stable
---
apiVersion: helm.toolkit.fluxcd.io/v2beta1
kind: HelmRelease
metadata:
  name: dynatrace-operator
  namespace: dynatrace
spec:
  interval: 5m
  chart:
    spec:
      chart: dynatrace-operator
      version: "1.2.x"  # SemVer range for auto-updates
      sourceRef:
        kind: HelmRepository
        name: dynatrace
        namespace: flux-system
  values:
    platform: kubernetes
    csidriver:
      enabled: true
```

Flux Kustomization for DynaKube

```yaml
apiVersion: kustomize.toolkit.fluxcd.io/v1
kind: Kustomization
metadata:
  name: dynakube
  namespace: flux-system
spec:
  interval: 10m
  path: ./overlays/prod
  prune: true
  sourceRef:
    kind: GitRepository
    name: monitoring-config
  healthChecks:
    - apiVersion: dynatrace.com/v1beta3
      kind: DynaKube
      name: dynakube
      namespace: dynatrace
  dependsOn:
    - name: dynatrace-operator
```

Flux GitRepository Source

```yaml
apiVersion: source.toolkit.fluxcd.io/v1
kind: GitRepository
metadata:
  name: monitoring-config
  namespace: flux-system
spec:
  interval: 1m
  url: https://github.com/your-org/monitoring-config
  ref:
    branch: main
  secretRef:
    name: git-credentials
```

4. Secret Management

API tokens should never be stored in Git. Use these approaches:

Option 1: External Secrets Operator

```yaml
apiVersion: external-secrets.io/v1beta1
kind: ExternalSecret
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  refreshInterval: 1h
  secretStoreRef:
    kind: ClusterSecretStore
    name: vault  # or aws-secrets-manager, gcp-secret-manager
  target:
    name: dynakube
    creationPolicy: Owner
  data:
    - secretKey: apiToken
      remoteRef:
        key: dynatrace/operator-token
    - secretKey: dataIngestToken
      remoteRef:
        key: dynatrace/data-ingest-token
```

Option 2: Sealed Secrets

```bash
# Create sealed secret from raw secret
kubectl create secret generic dynakube \
  --namespace dynatrace \
  --from-literal=apiToken=&lt;your-api-token&gt; \
  --from-literal=dataIngestToken=&lt;your-data-ingest-token&gt; \
  --dry-run=client -o yaml | \
  kubeseal --format yaml &gt; dynakube-sealed.yaml
```

```yaml
# Sealed secret (safe to commit)
apiVersion: bitnami.com/v1alpha1
kind: SealedSecret
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  encryptedData:
    apiToken: AgBy8hM...
    dataIngestToken: AgCtr9...
```

Option 3: SOPS with Age/GPG

```yaml
# .sops.yaml in repo root
creation_rules:
  - path_regex: .*.secret.yaml$
    age: age1ql3z7hjy54pw...
```

```bash
# Encrypt secrets
sops --encrypt secrets.yaml &gt; secrets.secret.yaml

# Flux decrypts automatically with kustomize-controller
```

5. Multi-Cluster Patterns

Kustomize Overlays by Environment

Base DynaKube (`base/dynakube/dynakube.yaml`):

```yaml
apiVersion: dynatrace.com/v1beta3
kind: DynaKube
metadata:
  name: dynakube
spec:
  apiUrl: PLACEHOLDER  # Patched per environment
  oneAgent:
    cloudNativeFullStack: {}
  activeGate:
    capabilities:
      - kubernetes-monitoring
      - routing
```

Production Overlay (`overlays/prod/kustomization.yaml`):

```yaml
apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
namespace: dynatrace
resources:
  - ../../base/dynakube
patches:
  - patch: |-
      - op: replace
        path: /spec/apiUrl
        value: https://your-tenant.live.dynatrace.com/api
    target:
      kind: DynaKube
      name: dynakube
```

Environment-Specific Configuration

Environment	Config Difference
Dev	Lower resources, fewer replicas
Staging	Match prod config, different tenant
Prod	High availability, more resources

Dev Patch (`overlays/dev/dynakube-patch.yaml`):

```yaml
apiVersion: dynatrace.com/v1beta3
kind: DynaKube
metadata:
  name: dynakube
spec:
  activeGate:
    replicas: 1
    resources:
      requests:
        cpu: 100m
        memory: 256Mi
```

6. Progressive Rollouts

ArgoCD Sync Waves

Control deployment order:

```
# Wave 0: CRDs and Namespace
apiVersion: v1
kind: Namespace
metadata:
  name: dynatrace
  annotations:
    argocd.argoproj.io/sync-wave: "0"
---
# Wave 1: Operator
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: dynatrace-operator
  annotations:
    argocd.argoproj.io/sync-wave: "1"
---
# Wave 2: DynaKube
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: dynakube
  annotations:
    argocd.argoproj.io/sync-wave: "2"
```

Flux Dependency Ordering

```
apiVersion: kustomize.toolkit.fluxcd.io/v1
kind: Kustomization
metadata:
  name: dynakube
spec:
  dependsOn:
    - name: dynatrace-operator
    - name: external-secrets  # If using ESO for tokens
```

Canary Deployments

Roll out monitoring changes gradually:

1. Deploy to dev cluster → validate
2. Deploy to staging → run tests
3. Deploy to prod-canary (subset) → monitor
4. Deploy to prod (all clusters)

7. Troubleshooting GitOps Deployments

ArgoCD Troubleshooting

```
# Check application status
argocd app get dynakube

# View sync diff
argocd app diff dynakube

# Force sync
argocd app sync dynakube --force

# View logs
kubectl -n argocd logs -l app.kubernetes.io/name=argocd-application-
controller
```

Flux Troubleshooting

```
# Check kustomization status
flux get kustomizations

# Check helm releases
flux get helmreleases -n dynatrace

# Force reconciliation
flux reconcile kustomization dynakube --with-source

# View events
kubectl -n flux-system get events --sort-by='.lastTimestamp'
```

Common Issues

Issue	Cause	Solution
Secret not found	External secret sync delay	Check ESO status, wait for sync
CRD not found	Operator not deployed	Check operator app status
Sync failed	YAML syntax error	Validate with `kubectl --dry-run`
Drift detected	Manual changes	Revert or accept drift in Git

```
// Monitor GitOps-related events in the cluster
fetch logs, from:-1h
| filter matchesPhrase(content, "argocd") or matchesPhrase(content, "flux")
| fields timestamp, content
| sort timestamp desc
| limit 30
```

```
// Track DynaKube configuration changes
fetch logs, from:-1h
| filter matchesPhrase(content, "dynakube") and (matchesPhrase(content,
"updated") or matchesPhrase(content, "created") or matchesPhrase(content,
"deleted"))
| fields timestamp, content
| sort timestamp desc
| limit 20
```

Next Steps

With GitOps configured, proceed to:

Next Notebook	Topic
K8S-04: Cluster Health Monitoring	Deep-dive into cluster metrics
K8S-05: Workload Monitoring	Application-level observability
K8S-06: Namespace Organization	Boundaries and access control

Summary

In this notebook, you learned:

- GitOps principles and benefits for monitoring
- ArgoCD Application definitions for operator and DynaKube
- Flux HelmRelease and Kustomization for DynaKube
- Secret management with External Secrets, Sealed Secrets, and SOPS
- Multi-cluster patterns with Kustomize overlays
- Progressive rollout strategies
- Troubleshooting GitOps deployments

References

- ArgoCD Documentation
- Flux Documentation
- External Secrets Operator
- Kustomize
